HHR与Depression的横断面相关性分析

基于NHANES与CHARLS的双库验证－cute_ran老师

中期分析结果

基于NHANES与CHARLS公共数据库

研究指标： HHR

研究人群： 成年社区/队列参与者（NHANES 与 CHARLS）

数据来源： NHANES、CHARLS

研究结局： Depression

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本报告为中期分析结果，包含主要统计分析内容及图表。

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1 研究背景与数据来源

本研究在 NHANES（2005–2023）与 CHARLS（2015）中评估 HHR 与 Depression 的横断面关联，并完成双库验证。

暴露指标为 HHR，计算公式为：

$$HHR = Hematocrit / Hemoglobin$$

1.1 纳排流程

图1给出两库纳排步骤与最终分析人数。

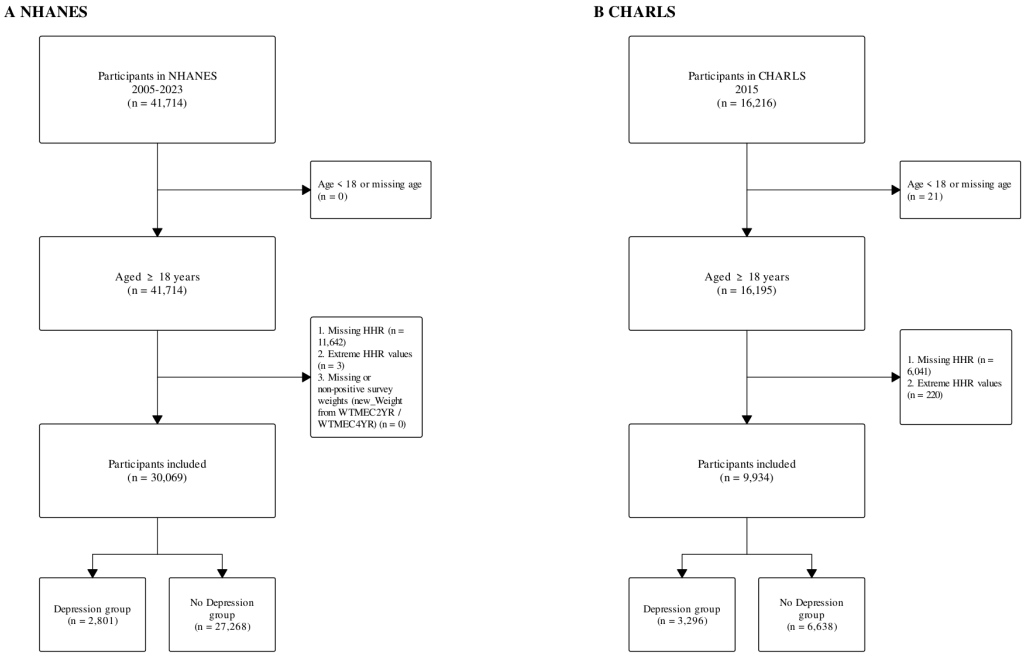


图1：研究对象纳排流程

注释：NHANES 起始 41,714 人，剔除缺失或极端 HHR 后纳入 30,069 人（Depression 2,801；No Depression 27,268）。CHARLS 起始 16,216 人，年龄缺失 21 人，剔除缺失或极端 HHR 后纳入 9,934 人（Depression 3,296；No Depression 6,638）。

2 分析方法

NHANES 采用复杂抽样设计加权 logistic；CHARLS 为非加权分析。协变量经单因素、VIF 与双库协调后，Model 1 调整 Age，Model 2 调整 Age、DBP、Hypertension、WBC。

缺失值以 MICE 多重插补（m=5），回归估计按 Rubin 规则合并；描述表与作图使用第 1 套完整数据。连续暴露另做 RCS；亚组在全分析集上比较最高与最低四分位；中介指标锁定 CRP。

3 基线特征

3.1 NHANES 基线

表1为 NHANES 加权基线特征（未加权人数）。

表1: NHANES 队列按 Depression 分层的基线特征

Characteristic	Overall (Unweighted N = 30069)	No Depression (Unweighted N = 27268)	Depression (Unweighted N = 2801)	p-value
Demographics				
Age,years	46.00 (32.00, 60.00)	46.00 (32.00, 61.00)	45.00 (31.00, 57.00)	<0.001
Gender				<0.001
Female	15,499.00 (51.26%)	13,741.00 (50.25%)	1,758.00 (62.47%)	
Male	14,570.00 (48.74%)	13,527.00 (49.75%)	1,043.00 (37.53%)	
Education				<0.001
Non-higher education	13,759.00 (39.00%)	12,228.00 (38.07%)	1,531.00 (49.22%)	
Higher education	16,310.00 (61.00%)	15,040.00 (61.93%)	1,270.00 (50.78%)	
Marital Status				<0.001
Unmarried	13,005.00 (40.07%)	11,504.00 (38.97%)	1,501.00 (52.28%)	
Married	17,064.00 (59.93%)	15,764.00 (61.03%)	1,300.00 (47.72%)	
Alcohol drinking				0.699
No	27,236.00 (91.85%)	24,700.00 (91.87%)	2,536.00 (91.64%)	
Yes	2,833.00 (8.15%)	2,568.00 (8.13%)	265.00 (8.36%)	
Weight,kg	80.00 (67.60, 95.10)	80.00 (67.50, 94.80)	81.00 (68.40, 99.80)	0.007
Height,cm	168.30 (161.20, 176.00)	168.50 (161.40, 176.20)	166.10 (159.40, 173.70)	<0.001
Waist circumference,cm	97.90 (87.00, 109.50)	97.70 (87.00, 109.10)	100.40 (88.20, 114.30)	<0.001
Vital Signs				
PP,mmHg	48.60 (40.00, 59.30)	48.70 (40.60, 59.30)	48.00 (40.00, 58.60)	0.002
SBP,mmHg	120.00 (110.70, 131.30)	120.00 (110.70, 131.30)	118.70 (109.30, 131.30)	0.091
DBP,mmHg	70.70 (63.30, 78.00)	70.70 (63.30, 78.00)	70.70 (63.30, 78.70)	0.368

Characteristic	Overall (Unweighted N = 30069)	No Depression (Unweighted N = 27268)	Depression (Unweighted N = 2801)	p-value
Laboratory Tests				
WBC,K/uL	6.90 (5.80, 8.40)	6.90 (5.70, 8.30)	7.40 (6.10, 9.00)	<0.001
BUN,mg/dL	13.00 (10.00, 16.00)	13.00 (10.00, 16.00)	12.00 (9.00, 15.00)	<0.001
Creatinine,mg/dL	0.85 (0.72, 1.00)	0.86 (0.72, 1.00)	0.82 (0.70, 0.96)	<0.001
Uric Acid,mg/dL	5.30 (4.40, 6.30)	5.30 (4.40, 6.30)	5.20 (4.30, 6.20)	<0.001
HbA1c,%	5.40 (5.20, 5.70)	5.40 (5.20, 5.70)	5.50 (5.20, 5.80)	0.005
Total Cholesterol,mg/dL	189.00 (164.00, 218.00)	190.00 (164.00, 218.00)	188.00 (161.00, 220.00)	0.338
Triglycerides,mg/dL	119.00 (80.00, 182.00)	118.00 (80.00, 181.00)	129.00 (82.00, 195.00)	<0.001
CRP,mg/L	1.80 (0.70, 4.40)	1.80 (0.70, 4.20)	2.40 (0.90, 6.10)	<0.001
Comorbidities				
Hypertension				<0.001
No	19,760.00 (69.46%)	18,147.00 (70.24%)	1,613.00 (60.88%)	
Yes	10,309.00 (30.54%)	9,121.00 (29.76%)	1,188.00 (39.12%)	
Diabetes				<0.001
No	26,290.00 (90.38%)	24,008.00 (90.84%)	2,282.00 (85.37%)	
Yes	3,779.00 (9.62%)	3,260.00 (9.16%)	519.00 (14.63%)	
Exposure				
HHR	2.95 (2.90, 3.00)	2.95 (2.89, 3.00)	2.96 (2.90, 3.02)	<0.001
Continuous variables are presented as weighted mean (standard error) or weighted median (interquartile range) depending on their distribution. Categorical variables are presented as weighted count (weighted percentage).				
Statistical comparisons were performed using survey-weighted Wilcoxon rank-sum tests and survey-adjusted chi-squared tests, accounting for the complex survey design (stratification, clustering, and sampling weights).				

注释：分析集未加权 N=30,069，其中 Depression 2,801、No Depression 27,268。Depression 组女性比例更高（62.47% vs 50.25%），CRP 中位数更高（2.40 vs 1.80 mg/L）。

3.2 CHARLS 基线

表2为 CHARLS 基线特征。

表2: CHARLS 队列按 Depression 分层的基线特征

Characteristic	Overall N = 9,934	No Depression N = 6,638	Depression N = 3,296	p-value
Demographics				
Age,years	59.00 (52.00, 66.00)	59.00 (51.00, 66.00)	60.00 (52.00, 67.00)	<0.001
Gender				<0.001
Female	5,326 (53.61%)	3,223 (48.55%)	2,103 (63.80%)	
Male	4,608 (46.39%)	3,415 (51.45%)	1,193 (36.20%)	
Education				0.016
Below High school	2,060 (20.74%)	1,330 (20.04%)	730 (22.15%)	
Above High school	7,874 (79.26%)	5,308 (79.96%)	2,566 (77.85%)	
Marital Status				<0.001
Unmarried	1,141 (11.49%)	633 (9.54%)	508 (15.41%)	
Married	8,793 (88.51%)	6,005 (90.46%)	2,788 (84.59%)	
Alcohol drinking				<0.001
No	6,337 (63.79%)	4,030 (60.71%)	2,307 (69.99%)	
Yes	3,597 (36.21%)	2,608 (39.29%)	989 (30.01%)	
Weight,kg	59.80 (52.70, 67.60)	60.90 (53.70, 68.60)	57.70 (51.00, 65.30)	<0.001
Height,cm	158.00 (152.50, 164.50)	159.20 (153.40, 165.50)	156.10 (151.00, 162.00)	<0.001
Waist circumference,cm	86.40 (79.30, 93.50)	86.80 (79.90, 94.00)	85.30 (78.50, 92.60)	<0.001
Vital Signs				
PP,mmHg	51.00 (43.00, 62.00)	51.00 (43.00, 62.00)	51.00 (43.00, 62.00)	0.908
SBP,mmHg	128.00 (116.00, 144.00)	129.00 (116.00, 144.00)	128.00 (114.00, 144.00)	0.033
DBP,mmHg	76.00 (69.00, 84.00)	76.00 (69.00, 84.00)	75.00 (68.00, 84.00)	<0.001
Laboratory Tests				
WBC,K/uL	5.70 (4.73, 6.80)	5.72 (4.80, 6.84)	5.60 (4.68, 6.80)	<0.001
BUN,mg/dL	14.57 (12.32, 17.65)	14.57 (12.32, 17.65)	14.57 (12.04, 17.65)	0.998
Creatinine,mg/dL	0.76 (0.65, 0.89)	0.77 (0.66, 0.90)	0.73 (0.64, 0.86)	<0.001
Uric Acid,mg/dL	4.80 (3.90, 5.70)	4.90 (4.00, 5.80)	4.60 (3.80, 5.50)	<0.001
Glucose,mg/dL	93.69 (88.29, 102.70)	93.69 (88.29, 102.70)	93.69 (86.49, 102.70)	0.028
HbA1c,%	5.80 (5.50, 6.10)	5.80 (5.50, 6.10)	5.80 (5.50, 6.10)	0.543
Total Cholesterol,mg/dL	181.08 (159.46, 205.41)	180.89 (159.46, 205.02)	181.85 (159.85, 206.18)	0.119
Triglycerides,mg/dL	111.50 (81.42, 162.83)	111.50 (81.42, 163.72)	110.62 (81.42, 161.06)	0.712
CRP,mg/L	1.30 (0.70, 2.50)	1.30 (0.70, 2.40)	1.40 (0.70, 2.60)	0.003
Comorbidities				

Characteristic	Overall N = 9,934	No Depression N = 6,638	Depression N = 3,296	p-value
Hypertension				0.112
No	6,812 (68.57%)	4,587 (69.10%)	2,225 (67.51%)	
Yes	3,122 (31.43%)	2,051 (30.90%)	1,071 (32.49%)	
Diabetes				0.002
No	8,127 (81.81%)	5,487 (82.66%)	2,640 (80.10%)	
Yes	1,807 (18.19%)	1,151 (17.34%)	656 (19.90%)	
Exposure				
HHR	3.038 (2.929, 3.151)	3.03 (2.93, 3.14)	3.05 (2.94, 3.17)	<0.001
Continuous variables are presented as mean (standard deviation) or median (interquartile range) depending on their distribution. Categorical variables are presented as count (percentage).				
Statistical comparisons were performed using the Wilcoxon rank-sum test or Pearson's chi-squared test.				

注释：分析集 N=9,934，Depression 3,296、No Depression 6,638。Depression 组女性占比 63.80%，高于非抑郁组 48.55%；年龄中位数分别为 60 与 59 岁。

4 主要关联结果

4.1 NHANES logistic

表3为 NHANES 加权 logistic 结果。

表3: NHANES 中 HHR 与 Depression 的加权 logistic 回归

				Crude Model			Model1			Model2	
Characteristic	Exposure cutoff	Events / N (%)	OR	95%CI	P-value	OR	95%CI	P-value	OR	95%CI	P-value
HHR											
HHR continuous			3.752	(1.885,7.471)	<0.001	4.164	(2.107,8.228)	<0.001	3.65	(1.867,7.136)	<0.001
HHR groups											
Q1 (Ref)	< 2.90	637/7,110 (8.959%)	Ref	Ref		Ref	Ref		Ref	Ref	
Q2	2.90 -< 2.95	600/7,167 (8.372%)	0.988	(0.837,1.166)	0.884	0.999	(0.845,1.181)	0.989	1.013	(0.858,1.195)	0.883
Q3	2.95 -< 3.00	613/6,635 (9.239%)	1.075	(0.905,1.277)	0.410	1.103	(0.929,1.309)	0.268	1.115	(0.941,1.321)	0.213
Q4	≥ 3.00	951/9,157 (10.385%)	1.293	(1.089,1.536)	0.004	1.335	(1.124,1.586)	0.001	1.306	(1.103,1.548)	0.003
p for trend					0.002			<0.001			0.001
The Crude Model was non-adjusted.											
The Model 1 was adjusted by: Age.											
The Model 2 was adjusted by: Age, DBP, Hypertension, WBC.											

注释：Model 2 下连续 HHR 的 OR=3.650（95%CI 1.867–7.136，P<0.001）；Q4 相对 Q1 的 OR=1.306（95%CI 1.103–1.548，P=0.003），趋势 P=0.001。

4.2 CHARLS logistic

表4为 CHARLS logistic 结果。

表4: CHARLS 中 HHR 与 Depression 的 logistic 回归

				Crude Model			Model1			Model2	
Characteristic	Exposure cutoff	Events / N (%)	OR	95%CI	P-value	OR	95%CI	P-value	OR	95%CI	P-value
HHR											
HHR continuous			1.449	(1.186, 1.77)	<0.001	1.424	(1.165, 1.74)	<0.001	1.388	(1.135, 1.698)	0.001
HHR groups											
Q1 (Ref)	< 2.93	789/2,483 (31.776%)	Ref	Ref		Ref	Ref		Ref	Ref	
Q2	2.93 -< 3.04	736/2,480 (29.677%)	0.906	(0.803, 1.022)	0.109	0.904	(0.801, 1.02)	0.100	0.902	(0.799, 1.018)	0.094
Q3	3.04 -< 3.15	837/2,483 (33.709%)	1.092	(0.97, 1.229)	0.147	1.081	(0.96, 1.218)	0.198	1.073	(0.952, 1.208)	0.248
Q4	≥ 3.15	934/2,488 (37.54%)	1.29	(1.148, 1.451)	<0.001	1.277	(1.136, 1.436)	<0.001	1.263	(1.123, 1.421)	<0.001
p for trend					<0.001			<0.001			<0.001
The Crude Model was non-adjusted.											
The Model 1 was adjusted by: Age.											
The Model 2 was adjusted by: Age, DBP, Hypertension, WBC.											
Multiple imputation by chained equations (m = 5, maxit = 5, seed = 1234).											
Regression estimates (Cox/logistic) were pooled across imputations using Rubin's rules.											
Operational steps (feature selection, KM/RCS plots, descriptive tables) used the first completed dataset.											

注释：Model 2 下连续 HHR 的 OR=1.388（95%CI 1.135－1.698，P=0.001）；Q4 相对 Q1 的 OR=1.263（95%CI 1.123－1.421，P<0.001），趋势 P<0.001。

4.3 剂量反应 RCS

图2为双库 RCS 曲线。

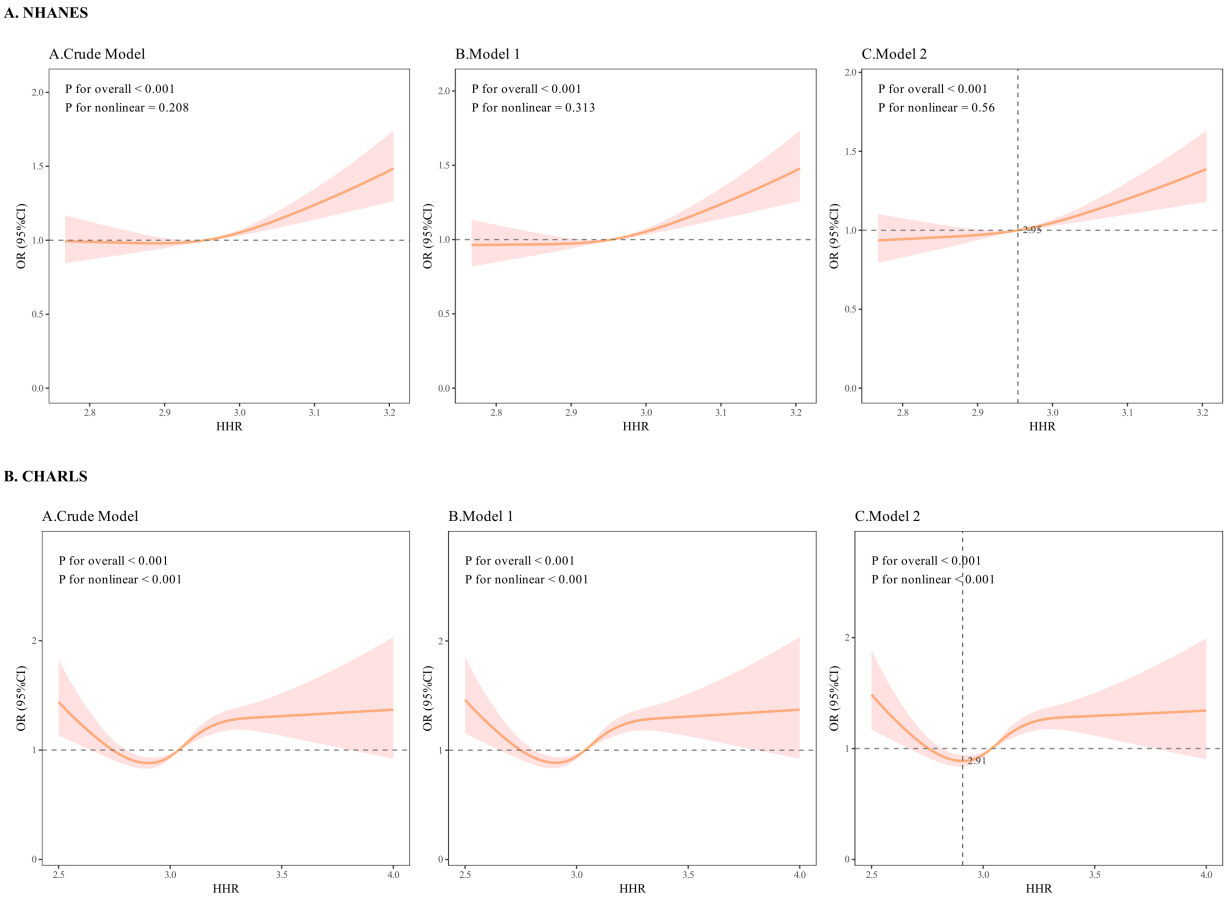


图2：HHR 与 Depression 的限制性立方样条

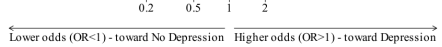
注释：两库 Model 2 的总体关联均显著（P for overall <0.001）。NHANES 非线性不显著（P for nonlinear=0.560）；CHARLS 非线性显著（P for nonlinear <0.001）。

4.4 亚组分析

图3为双库亚组森林图（最高 vs 最低四分位）。

A. NHANES

Subgroup	N	OR (95% CI)	P value	P for interaction
Age Group				**0.884**
< 65	22698	1.31 (1.08-1.59)	0.008	
≥ 65	7371	1.81 (1.26-2.58)	0.002	
Gender				**0.140**
Female	15499	1.12 (0.91-1.38)	0.268	
Male	14570	1.34 (1.03-1.74)	0.032	
Education				**0.812**
Higher education	16310	1.35 (1.06-1.72)	0.017	
Non-higher education	13759	1.28 (1.04-1.59)	0.025	
Marital Status				**0.742**
Unmarried	13005	1.19 (0.97-1.46)	0.101	
Married	17064	1.35 (1.04-1.76)	0.026	
Alcohol drinking				**0.243**
No	27236	1.39 (1.16-1.67)	<0.001	
Yes	2833	0.81 (0.50-1.31)	0.401	
Hypertension				**0.297**
No	19760	1.23 (1.00-1.50)	0.053	
Yes	10309	1.39 (1.04-1.85)	0.028	



B. CHARLS

Subgroup	N	OR (95% CI)	P value	P for interaction
Age Group				**0.247**
< 65	6879	1.36 (1.18-1.57)	<0.001	
≥ 65	3055	1.12 (0.91-1.37)	0.298	
Gender				**0.632**
Female	5326	1.14 (0.97-1.33)	0.101	
Male	4608	1.23 (1.02-1.48)	0.027	
Education				**0.856**
Above High school	7874	1.30 (1.14-1.48)	<0.001	
Below High school	2060	1.25 (0.97-1.62)	0.089	
Marital Status				**0.789**
Unmarried	1141	1.17 (0.84-1.62)	0.36	
Married	8793	1.31 (1.15-1.48)	<0.001	
Alcohol drinking				**0.281**
No	6337	1.18 (1.02-1.36)	0.026	
Yes	3597	1.44 (1.18-1.76)	<0.001	
Hypertension				**0.365**
No	6812	1.36 (1.18-1.57)	<0.001	
Yes	3122	1.16 (0.95-1.43)	0.152	

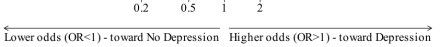


图3: HHR 与 Depression 的亚组分析

注释：NHANES 与 CHARLS 各亚组 Age 分层人数之和分别等于 30,069 与 9,934。多数分层 OR>1；两库各亚组交互 P 均>0.05，未见明确效应修饰。

5 补充分析

5.1 分布比较

图4按 Depression 分组比较 HHR 分布。

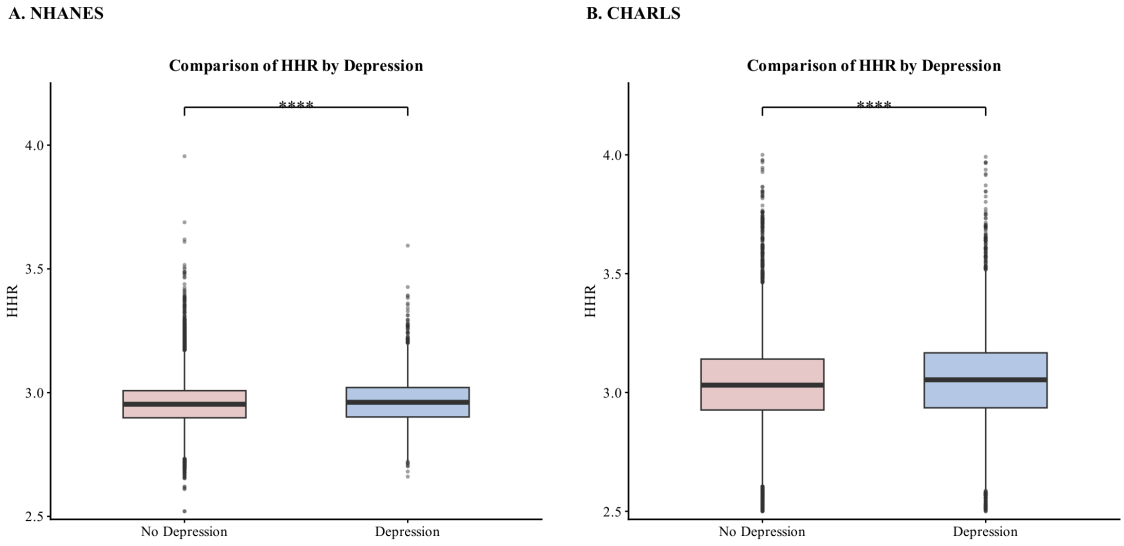


图4: 按 Depression 分组的 HHR 箱线图

注释：两库 Depression 与 No Depression 组 HHR 分布均可比较；组间差异检验见原图标注。

5.2 判别能力

图5为多变量 ROC。

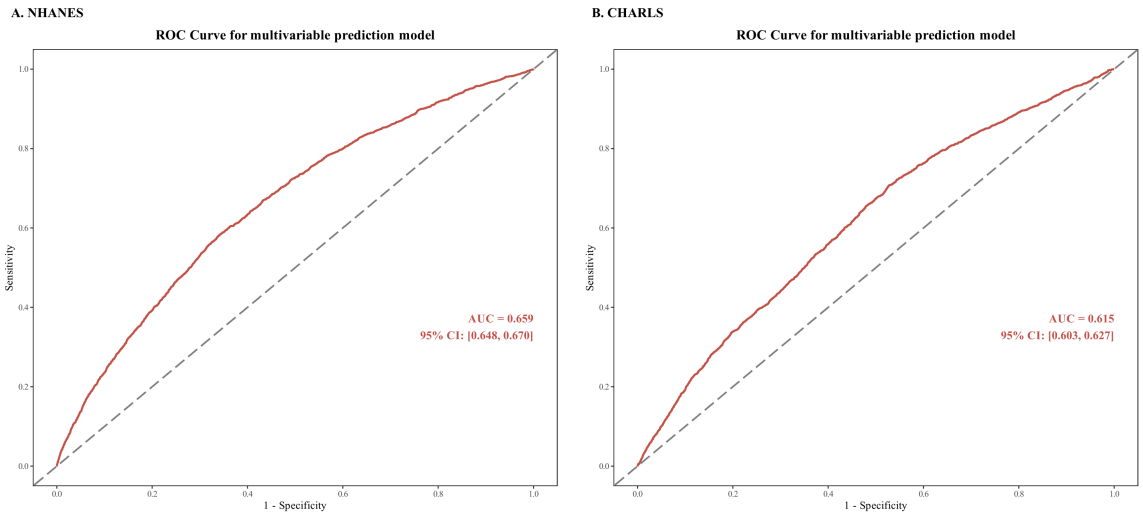


图5：多变量模型 ROC

注释：NHANES AUC=0.659（95%CI 0.648–0.670）；CHARLS AUC=0.615（95%CI 0.603–0.627），判别能力中等偏弱。

5.3 中介分析

表5、表6与图6给出 CRP 中介结果。

表5：NHANES 中 HHR 经 CRP 的中介分析

Mediator	Total Effect (Index)	Mediator–outcome OR(1-SD)	Direct Effect	Indirect Effect	Path a (Beta)	Path b (Beta)	Proportion mediated
CRP	3.217 [1.612–6.42] 0.001	1.128 [1.088–1.171] < 0.001	2.99 [1.5–5.958] 0.003	1.057 [1.013–1.103] 0.011	8.783 [6.854, 10.711] < 0.001	0.006 [0.002, 0.011] 0.009	4.760%
A negative proportion mediated indicates a suppression (masking) effect, not a mediated fraction.							

注释：中介为 CRP。NHANES Direct Effect=2.99（1.5–5.958），P=0.003，Proportion mediated=4.760%。

5.4 CHARLS 中介

CHARLS 中介表如下。

表6：CHARLS 中 HHR 经 CRP 的中介分析

Mediator	Total Effect (Index)	Mediator-outcome OR(1-SD)	Direct Effect	Indirect Effect	Path a (Beta)	Path b (Beta)	Proportion mediated
CRP	1.404 [1.148-1.716] < 0.001	1.053 [1.012-1.097] 0.011	1.394 [1.14-1.704] 0.001	1.007 [1-1.014] 0.065	0.113 [0.02, 0.206] 0.017	0.061 [0.019, 0.102] 0.004 (per 1-SD M)	2.020%
Path coefficients and mediator-outcome effects are per 1-SD of the mediator.							
A negative proportion mediated indicates a suppression (masking) effect, not a mediated fraction.							

注释：CHARLS Direct Effect=1.394（1.14-1.704），P=0.001，Proportion mediated=2.020%。

5.5 中介路径图

图6为双库中介路径图。

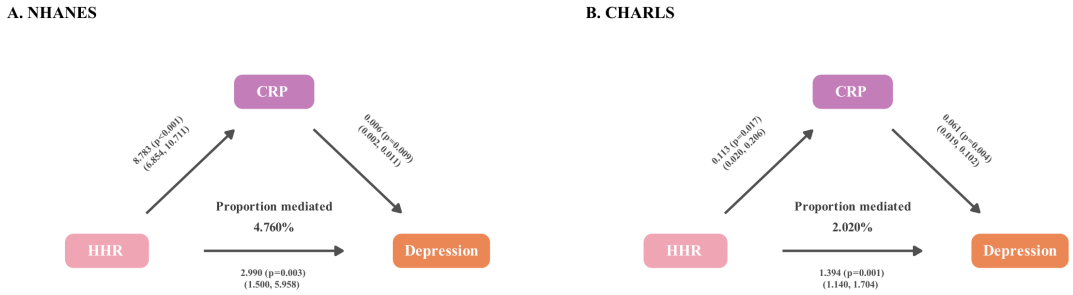


图6：HHR－CRP－Depression 中介路径图

注释：左右面板分别为 NHANES 与 CHARLS；路径系数与 Table S8 中 Direct Effect、Proportion mediated 一致。

5.6 排除饮酒敏感性

表7、表8为排除饮酒者后的关联结果。

表7: NHANES 排除饮酒后的 logistic 敏感性分析

				Crude Model			Model1			Model2	
Characteristic	Exposure cutoff	Events / N (%)	OR	95%CI	P-value	OR	95%CI	P-value	OR	95%CI	P-value
HHR											
HHR continuous			4.477	(2.291,8.748)	<0.001	5.01	(2.584,9.71)	<0.001	4.363	(2.254,8.445)	<0.001
HHR groups											
Q1 (Ref)	< 2.89	568/6,436 (8.825%)	Ref	Ref		Ref	Ref		Ref	Ref	
Q2	2.89 -< 2.95	533/6,526 (8.167%)	0.989	(0.832,1.176)	0.904	1.001	(0.84,1.193)	0.992	1.013	(0.851,1.206)	0.883
Q3	2.95 -< 3.00	563/6,107 (9.219%)	1.099	(0.924,1.306)	0.290	1.129	(0.95,1.341)	0.173	1.139	(0.959,1.352)	0.141
Q4	≥ 3.00	872/8,167 (10.677%)	1.335	(1.121,1.59)	0.002	1.381	(1.159,1.646)	<0.001	1.35	(1.135,1.606)	0.001
p for trend					<0.001			<0.001			<0.001
The Crude Model was non-adjusted.											
The Model 1 was adjusted by: Age.											
The Model 2 was adjusted by: Age, DBP, Hypertension, WBC.											

注释：排除饮酒后关联方向与主分析一致，具体 OR 见原表 Model 2。

5.7 CHARLS 排除饮酒

CHARLS 排除饮酒关联表如下。

表8: CHARLS 排除饮酒后的 logistic 敏感性分析

				Crude Model			Model1			Model2	
Characteristic	Exposure cutoff	Events / N (%)	OR	95%CI	P-value	OR	95%CI	P-value	OR	95%CI	P-value
HHR											
HHR continuous			1.29	(1.007,1.652)	0.044	1.272	(0.993,1.631)	0.057	1.265	(0.986,1.622)	0.064
HHR groups											
Q1 (Ref)	< 2.94	559/1,580 (35.38%)	Ref	Ref		Ref	Ref		Ref	Ref	
Q2	2.94 -< 3.05	527/1,586 (33.228%)	0.909	(0.785,1.053)	0.202	0.906	(0.782,1.05)	0.188	0.908	(0.784,1.052)	0.200
Q3	3.05 -< 3.16	595/1,586 (37.516%)	1.097	(0.949,1.267)	0.212	1.09	(0.943,1.26)	0.243	1.092	(0.944,1.262)	0.237
Q4	≥ 3.16	626/1,585 (39.495%)	1.192	(1.032,1.377)	0.017	1.183	(1.024,1.367)	0.022	1.183	(1.023,1.367)	0.023
p for trend					0.002			0.003			0.003
The Crude Model was non-adjusted.											
The Model 1 was adjusted by: Age.											
The Model 2 was adjusted by: Age, DBP, Hypertension, WBC.											
Multiple imputation by chained equations (m = 5, maxit = 5, seed = 1234).											
Regression estimates (Cox/logistic) were pooled across imputations using Rubin's rules.											
Operational steps (feature selection, KM/RCS plots, descriptive tables) used the first completed dataset.											

注释：CHARLS 排除饮酒后 Q4 vs Q1 关联仍为正向，详见原表。

6 小结

在 NHANES 与 CHARLS 中，较高 HHR 与 Depression 呈正向关联；四分位趋势与连续模型结果一致。

缩略词对照表

缩写词	对应英文	对应中文	单位
NHANES	National Health and Nutrition Examination Survey	美国国家健康与营养调查	
CHARLS	China Health and Retirement Longitudinal Study	中国健康与养老追踪调查	
HHR	Hematocrit-to-Hemoglobin Ratio	红细胞压积/血红蛋白比值	
OR	Odds Ratio	比值比	
CI	Confidence Interval	置信区间	
RCS	Restricted Cubic Spline	限制性立方样条	
ROC	Receiver Operating Characteristic	受试者工作特征	
AUC	Area Under the Curve	曲线下面积	
VIF	Variance Inflation Factor	方差膨胀因子	
MI	Multiple Imputation	多重插补	
CRP	C-reactive Protein	C反应蛋白	mg/L
WBC	White Blood Cell count	白细胞计数	K/uL
DBP	Diastolic Blood Pressure	舒张压	mmHg
Q1	Lowest quartile	最低四分位	
Q4	Highest quartile	最高四分位	

本报告为中期分析结果，最终结论以正式交付为准。

分析日期：2026年10月23日